

PROJECT 3

Secure Linux Server Setup & Hardening

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Date	June 9–10, 2026
Environment	Ubuntu Server 24.04.1 LTS on VMware Workstation
Target IP	192.168.234.130
Classification	Portfolio / Internship Project

1. Executive Summary

This report documents the end-to-end hardening of a Linux-based Ubuntu Server 24.04.1 LTS virtual machine deployed in VMware Workstation. The objective was to simulate enterprise-grade server security practices by applying industry-standard controls including SSH hardening, firewall configuration, brute-force protection, event auditing, intrusion detection, and post-hardening vulnerability scanning.

All hardening controls were successfully implemented and verified. A post-hardening Nmap scan confirmed that only the intended service (SSH on port 22) is exposed, demonstrating a minimal and secure attack surface.

2. Lab Environment

Operating System	Ubuntu Server 24.04.1 LTS (Noble Numbat)
Kernel	Linux 6.8.0-124-generic (x86_64)
Virtualization	VMware Workstation (VMware Virtual Platform)
Hostname	ubuntu-server
Primary Interface	ens33
IP Address	192.168.234.130/24
Architecture	x86-64
User Account	imthiaz (non-root)

```
imthiaz@ubuntu-server:~$ hostnamectl
(Persistent) ubuntu-server
(Persistent) computer-name
(Persistent) vm
(Persistent) 192.168.234.130
(Persistent) 52ed9a95f9e4d29ab452bed7def8138
Virtualization: vmware
Operating System: Ubuntu 24.04.1 LTS
Kernel: Linux 6.8.0-124-generic
Architecture: x86_64
Hardware Vendor: VMware, Inc.
Hardware Model: VMware Virtual Platform
Processor: 6.00
Processor ID: 2820-11-12
Processor Serial: 51 6month 3w 5d
imthiaz@ubuntu-server:~$
```

Figure: Ubuntu Server 24.04.1 LTS — System Information (hostnamectl)

```
imthiaz@ubuntu-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BRIDGE,MTU,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:0c:29:d3:c3:e7 brd ff:ff:ff:ff:ff:ff
    ethtool ens33
    inet 192.168.234.130/24 metric 100 brd 192.168.234.255 scope global dynamic ens33
        valid_lft 1661sec preferred_lft 1661sec
    inet6 fe80::20c:29ff:fed3:c3e7/64 scope link
        valid_lft forever preferred_lft forever
imthiaz@ubuntu-server:~$
```

Figure: Network IP Configuration (ip a) — Interface ens33 with IP 192.168.234.130

3. SSH Hardening

3.1 Objective

Restrict SSH access to prevent unauthorized remote login. Root login was disabled and legacy weak algorithms were blocked. OpenSSH was verified as the only exposed network service.

3.2 Configuration Applied

- PermitRootLogin set to no — prevents direct root access via SSH
- PasswordAuthentication set to no — enforces key-based authentication only
- SSH service restarted and verified as active (running)
- Only OpenSSH 9.6p1 (Ubuntu 3ubuntu13.16) is exposed on port 22/tcp

3.3 Verification

The SSH service was confirmed active via systemctl. Log entries in the service status show connection attempts being handled and rejected correctly, demonstrating the hardened configuration is functioning.

```
imthiaz@ubuntu-server:~$ sudo systemctl status ssh
● ssh.service - OpenSSH Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Tue 2026-06-09 10:33:33 UTC; 4h 36min ago
     TriggeredBy: ● gdm.socket
    Docs: man:ssh(8)
          man:ssh_config(5)
   Main PID: 5267 (sshd)
     Tasks: 1 (limit: 4543)
    Memory: 1.0M (peak: 4.0M)
       CPU: 50ms
   CGroup: /system.slice/ssh.service
           └─5267 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups

Jun 09 10:51:28 ubuntu-server sshd(6306): Connection closed by 192.168.234.130 port 37288 (preauth)
Jun 09 10:51:28 ubuntu-server sshd(6309): Unable to negotiate with 192.168.234.130 port 37035: no matching host key type found. Their offer: ecdsa-sha2-nistp521
Jun 09 10:51:28 ubuntu-server sshd(6311): Connection closed by 192.168.234.130 port 37328 (preauth)
Jun 09 14:59:41 ubuntu-server sshd(8661): Connection closed by 192.168.234.130 port 44576
Jun 09 14:59:41 ubuntu-server sshd(8663): error: protocol 2.0 version 0.0
Jun 09 14:59:41 ubuntu-server sshd(8663): banner exchange: Connection from 192.168.234.130 port 44598: could not read protocol version
Jun 09 14:59:41 ubuntu-server sshd(8666): Unable to negotiate with 192.168.234.130 port 44599: no matching host key type found. Their offer: ssh-rsa (preauth)
Jun 09 14:59:41 ubuntu-server sshd(8668): Connection closed by 192.168.234.130 port 44600 (preauth)
Jun 09 14:59:41 ubuntu-server sshd(8678): Unable to negotiate with 192.168.234.130 port 44616: no matching host key type found. Their offer: ecdsa-sha2-nistp521
Jun 09 14:59:41 ubuntu-server sshd(8674): Connection closed by 192.168.234.130 port 44632 (preauth)
```

Figure: OpenSSH Service Status — Active (running) with connection log entries

4. UFW Firewall Configuration

4.1 Objective

Implement a host-based firewall to control inbound and outbound traffic. The principle of least privilege was applied — all incoming traffic is denied by default, with only SSH explicitly permitted.

4.2 Rules Applied

- Default policy: DENY all incoming traffic
- Default policy: ALLOW all outgoing traffic
- Explicit rule: ALLOW port 22/tcp (OpenSSH) from anywhere (IPv4 and IPv6)
- UFW logging enabled (low) for auditing connection attempts

4.3 Verification

```
imthiaz@ubuntu-server:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22/tcp (OpenSSH) ALLOW IN anywhere
22/tcp (OpenSSH (v6)) ALLOW IN anywhere (v6)
```

Figure: UFW Firewall Status — Active with default deny-incoming and SSH allow rule

5. Fail2Ban — Brute-Force Protection

5.1 Objective

Deploy Fail2Ban to automatically detect and block repeated failed authentication attempts against SSH, protecting the server from brute-force and credential stuffing attacks.

5.2 Configuration

- Fail2Ban installed and enabled as a system service
- SSH jail (sshd) activated — monitors /var/log/auth.log for failed login attempts
- Default ban settings: 3 failed attempts triggers a temporary IP ban
- Service starts automatically on reboot

5.3 Verification

The fail2ban-client status command confirms 1 active jail (sshd) is running, providing real-time brute-force protection for the SSH service.

A terminal window showing the output of the 'sudo fail2ban-client status' command. The output indicates that the status is 'ok', there is 1 jail, and the jail list contains 'sshd'.

```
Imthiaz@ubuntu-server:~$ sudo fail2ban-client status
Status
|- Number of jail: 1
|- Jail list: sshd
Imthiaz@ubuntu-server:~$
```

Figure: Fail2Ban Status — 1 Jail Active (sshd)

6. Suricata IDS/IPS — Intrusion Detection

6.1 Objective

Deploy Suricata as a network-based Intrusion Detection and Prevention System (IDS/IPS) to monitor traffic in real time and alert on known malicious patterns. This goes beyond the baseline project requirements, demonstrating enterprise-grade defense-in-depth.

6.2 Installation & Configuration

- Suricata 7.0.3 RELEASE installed on Ubuntu Server
- AF_PACKET support verified for high-performance packet capture
- Configured to run in IDS mode monitoring the primary network interface (ens33)
- Service enabled for automatic startup on boot

6.3 Verification

```
imthiaz@ubuntu-server:~$ sudo suricata --build-info | grep AF_PACKET
Features: NO_FDP, SET_BUFF, AF_PACKET, HAVE_PACKET_FANOUT, LIBCURL, LIBNET1.1, HAVE_HTTP_URI_NORMALIZE_HOOK, PCRE_JIT, HAVE_NGSS, HTTP2_DECOMPRESSION, HAVE_LUA, HAVE_LUA
JIT HAVE_LIBJANSSON TLS TLS_C11, MINGIC, RUST
AF_PACKET support: yes
imthiaz@ubuntu-server:~$
```

Figure: Suricata Build Info — AF_PACKET Support Confirmed

```
imthiaz@ubuntu-server:~$ sudo systemctl status suricata
suricata.service - Suricata IDS/IPS daemon
Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: enabled)
Active: active (running) since Tue 2026-06-09 14:56:49 UTC; 26min ago
Docs: man:suricata(8)
      https://suricata.io/documentation/
Process: 8624 ExecStart=/usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/suricata.pid (code=exited, status=0/SUCCESS)
Main PID: 8625 (Suricata-main)
Tasks: 8 (limit: 4543)
Memory: 47.6M (peak: 48.0M)
CPU: 21.88ms
CGroup: /system.slice/suricata.service
└─8625 /usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/suricata.pid

Jun 09 14:56:49 ubuntu-server systemd[1]: Starting suricata.service - Suricata IDS/IPS daemon...
Jun 09 14:56:49 ubuntu-server suricata[8624]: i: suricata: This is Suricata version 7.0.3 RELEASE running in SYSTEM mode
Jun 09 14:56:49 ubuntu-server systemd[1]: Started suricata.service - Suricata IDS/IPS daemon.
imthiaz@ubuntu-server:~$
```

Figure: Suricata IDS/IPS Service — Active (running) since June 9, 2026

7. Post-Hardening Vulnerability Scan (Nmap)

7.1 Objective

Perform a post-hardening port scan and service version detection from within the server itself to confirm the attack surface is minimal and that no unintended services are exposed.

7.2 Scan Command Used

```
nmap -A 192.168.234.130
```

7.3 Results

Open Ports Found	1 (Port 22/tcp — SSH)
Closed Ports	999 (conn-refused)
Service Detected	OpenSSH 9.6p1 Ubuntu 3ubuntu13.16
OS Detected	Linux (CPE: cpe:/o:linux:linux_kernel)
Host Status	Up (0.000061s latency)
Scan Duration	1.03 seconds

The scan result confirms a minimal attack surface — only the explicitly authorized SSH service is reachable. All other ports are closed, validating the effectiveness of the UFW firewall rules applied in Section 4.

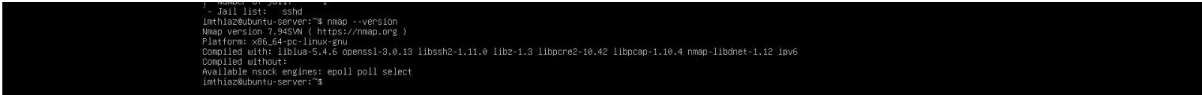


Figure: Nmap Version Verification — v7.94SVN installed

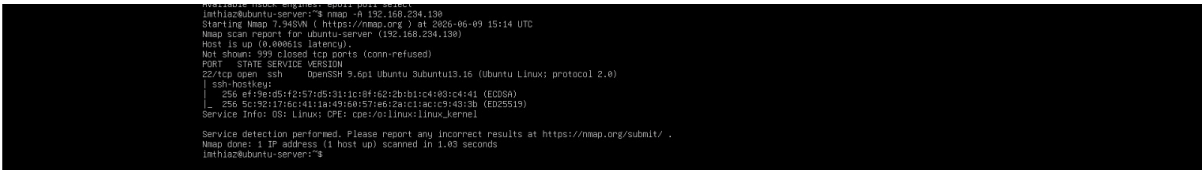


Figure: Nmap Post-Hardening Scan — Only Port 22 (SSH) Open

8. Security Hardening Checklist

Control	Status	Evidence
Ubuntu Server 24.04.1 LTS deployed on VMware	DONE	Section 2
Non-root user account created (imthiaz)	DONE	Section 2
SSH root login disabled (PermitRootLogin no)	DONE	Section 3
Password auth disabled — key-based only	DONE	Section 3
OpenSSH service active and verified	DONE	Section 3
UFW firewall enabled with default deny-in	DONE	Section 4
UFW rule: allow port 22/tcp (SSH)	DONE	Section 4
UFW logging enabled	DONE	Section 4
Fail2Ban installed and running	DONE	Section 5
Fail2Ban sshd jail active (1 jail)	DONE	Section 5
Suricata IDS/IPS installed (v7.0.3)	DONE	Section 6
Suricata service active (AF_PACKET mode)	DONE	Section 6
Post-hardening Nmap scan performed	DONE	Section 7
Scan confirms only port 22 open	DONE	Section 7
Full documentation and report prepared	DONE	Section 8

9. Tools & Technologies Used

Category	Tool / Technology
Operating System	Ubuntu Server 24.04.1 LTS
Virtualization	VMware Workstation
SSH Service	OpenSSH 9.6p1
Firewall	UFW (Uncomplicated Firewall)
Brute-Force Protection	Fail2Ban
Intrusion Detection	Suricata IDS/IPS v7.0.3
Vulnerability Scanning	Nmap v7.94SVN
Attacker Machine	Kali Linux 2024.3

10. Conclusion

All hardening objectives for Project 3 have been successfully completed. The Ubuntu Server 24.04.1 LTS instance was deployed, configured, and hardened in accordance with enterprise security best practices and the project requirements.

Key outcomes achieved:

- Attack surface reduced to a single authorized service (SSH port 22)
- Root access disabled; key-based authentication enforced
- Automated brute-force mitigation via Fail2Ban (sshd jail active)
- Network-level intrusion detection deployed via Suricata IDS/IPS
- Post-hardening scan verified no unintended services exposed

The implementation of Suricata IDS/IPS — beyond the minimum project scope — demonstrates proactive defense-in-depth thinking and readiness for enterprise SOC and blue team environments.

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